

BIJZLI CLASSES

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Dark Mode

BIJZLI Classes CS Test Series | Mock GATE | Test 1

SUBJECT Mock GATE

BATCH BIJZLI Classes Test Series for GATE CSE 2027

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RESULT SUMMARY

BIJZLI Classes CS Test Series | Mock GATE | Test 1 | Report

Time Used: 00:30:15

Marks Obtained
6.33 / 100

Total Marks
100

Positive Marks
7

Negative Marks
0.67

Total Questions
65

Attempted
7

Unattempted
58

Correct
6

Incorrect
1

Partial
0

Marked for Review
0

Accuracy
85.71%

Section-wise Analysis

SECTION	TOTAL	ATTEMPTED	CORRECT	INCORRECT	PARTIAL	SCORE
Technical	65	7	6	1	0	6.33

Question Review

All (65)

Correct (6)

Incorrect (1)

Partial (0)

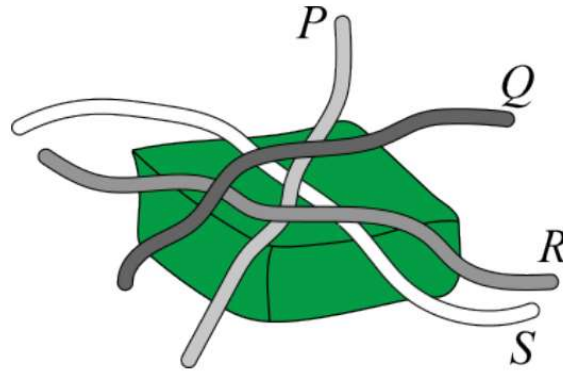
Unattempted (58)

Marked (0)

Q1 | Technical

CORRECT

Banh Chung is a traditional rice cake dish eaten in Vietnam to celebrate the Lunar New Year. These rice cakes are cut using strings placed on the cake one at a time and in different directions.
In what order were the four strings placed on the rice cake shown below?



Your Answer

D. S, P, R, Q

Correct Answer

D. S, P, R, Q **Option A** **S, P, Q, R** **Option B** **Q, S, P, R** **Option C** **R, P, S, Q** **Option D** **S, P, R, Q** **CORRECT** **YOUR CHOICE****OFFICIAL SOLUTION****Answer: D**[Official discussion page](#)**Q2 | Technical****CORRECT**The sum of $x\%$ of y , and $y\%$ of x is

Your Answer

A. 2% of xy

Correct Answer

A. 2% of xy **Option A** **2% of xy** **CORRECT** **YOUR CHOICE****Option B** **$(x + y)\%$ of xy** **Option C**

$(x + y)\%$ of 2**Option D** $xy\%$ of $(x + y)$ **OFFICIAL SOLUTION****Answer: A**[➤ Official discussion page](#)**Q3 | Technical****UNATTEMPTED**

For what values of ' k ' will the pair of equations $3x + 4y = 12$ and $kx + 12y = 30$ NOT have a unique solution?

Your Answer

Not answered

Correct Answer

C. 9**Option A****3****Option B****7.5****Option C****9****CORRECT****Option D****12****OFFICIAL SOLUTION****Answer: C**[➤ Official discussion page](#)**Q4 | Technical****CORRECT**

Bhavya sold her house at a loss of **25** percent of the price she originally paid for the house and then bought another house at a price of **30** percent less than the price she originally paid for her first house. If she sold the first house for **22, 50, 000**, what was her net gain, in rupees, for the two transactions?

Your Answer

B. 1, 50, 000

Correct Answer

B. 1, 50, 000**Option A**

1, 12, 500

Option B

CORRECT YOUR CHOICE

1, 50, 000

Option C

2, 50, 000

Option D

6, 00, 000

OFFICIAL SOLUTION

Answer: B

[↗ Official discussion page](#)

Q5 | Technical

CORRECT

The sight of a single actor portraying several characters in the same scene is no longer a shock to the average moviegoer, such special-effects trickery having become so _____.

Your Answer

D. commonplace

Correct Answer

D. commonplace

Option A

expensive

Option B

specialized

Option C

sinister

Option D

CORRECT YOUR CHOICE

commonplace

OFFICIAL SOLUTION

Answer: D

[↗ Official discussion page](#)



Q6 | Technical

CORRECT

Savanah tossed a fair coin some number of times, and **50%** of those tosses resulted in tails. She then tossed the coin one final time, and the result was tails. If **60%** of all tosses resulted in tails, how many tosses did she make in total?

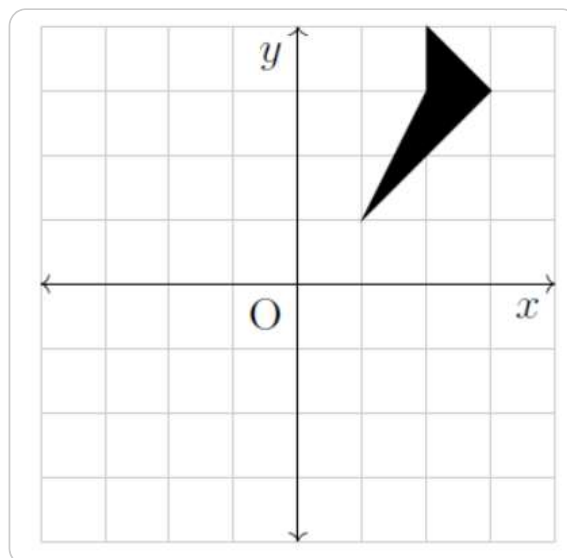
Your Answer

D. 5

Correct Answer

D. 5**Option A****3****Option B****4****Option C****8****Option D****5****CORRECT** **YOUR CHOICE****OFFICIAL SOLUTION****Answer: D**[↗ Official discussion page](#)**Q7 | Technical****INCORRECT**

If the shaded object below is first reflected about the x-axis and then rotated around the origin (O), it will now be seen as:



Your Answer

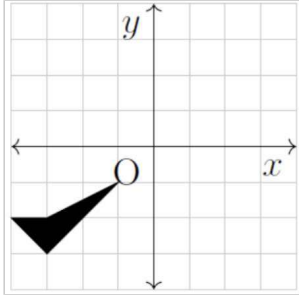
Correct Answer

B.

A.

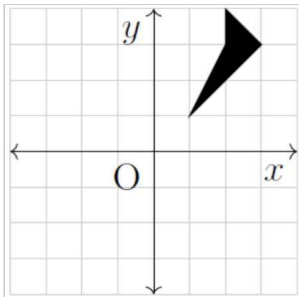
Option A

CORRECT

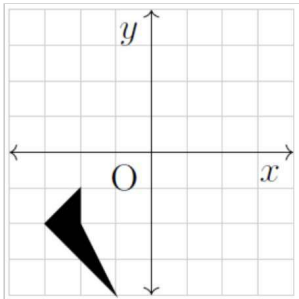


Option B

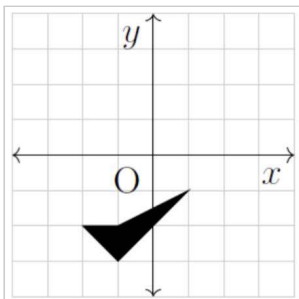
YOUR CHOICE



Option C



Option D



OFFICIAL SOLUTION

Answer: A

[Official discussion page](#)

If $\log_8 a + \log_4 b^2 = 5$ and $\log_8 b + \log_4 a^2 = 7$ then $a + b$ is

Your Answer

Not answered

Correct Answer

C. 72

Option A

64

Option B

68

Option C

72

CORRECT

Option D

80

OFFICIAL SOLUTION

Answer: C

[Official discussion page](#)

Q9 | Technical

UNATTEMPTED

A box contains **731** black balls and **2000** white balls. The following process is to be repeated as long as possible. Arbitrarily select two balls from the box. If they are of the same colour, throw them out and put a black ball into the box (enough extra black balls are available to do this). If they are of different colours, place the white ball back into the box and throw the black ball away. Which of the following is correct?

Your Answer

Not answered

Correct Answer

C. The process will stop with a single black ball in the box

Option A

The process can be applied indefinitely without any a priori bound

Option B

The process will stop with a single white ball in the box

Option C

The process will stop with a single black ball in the box

CORRECT

Option D

The process will stop with the box empty

OFFICIAL SOLUTION

Answer: C

[↗ Official discussion page](#)

Q10 | Technical

UNATTEMPTED

The first few positive whole numbers that are not powers of **3** are **2, 4, 5, 6, 7, 8, 10**. What is the sum of all the positive whole numbers that are less than **3^{10}** and are also not powers of **3**?

Your Answer

Not answered

Correct Answer

A. $\frac{(3^{10}-1)^2}{2}$

Option A

$$\frac{(3^{10}-1)^2}{2}$$

CORRECT

Option B

$$\frac{(3^{11}-1)^2}{2}$$

Option C

$$\frac{3(3^{10}-1)^2}{2}$$

Option D

$$(3^{10} - 1)^2$$

OFFICIAL SOLUTION

Answer: A

[↗ Official discussion page](#)



Q11 | Technical

CORRECT

According to De Morgan's theorem, the complement of **$X + Y \cdot Z$** is:

Your Answer

C. $\bar{X} \cdot (\bar{Y} + \bar{Z})$

Correct Answer

C. $\bar{X} \cdot (\bar{Y} + \bar{Z})$

Option A

$$\bar{X} + \bar{Y} \cdot \bar{Z}$$

Option B

$$\bar{X} \cdot \bar{Y} + \bar{Z}$$

Option C

CORRECT YOUR CHOICE

$$\bar{X} \cdot (\bar{Y} + \bar{Z})$$

Option D

$$\bar{X} \cdot \bar{Y} \cdot \bar{Z}$$

OFFICIAL SOLUTION

Answer: C

[↗ Official discussion page](#)

Q12 | Technical

UNATTEMPTED

Consider the following pseudocode:

```

SillySort( $A$ ) :
     $n \leftarrow \text{length}(A)$ 
    if ( $n = 0$  or  $n = 1$ ) :
        return  $A$ 
    left  $\leftarrow$  SillySort( $A[1 \dots \lfloor n/2 \rfloor]$ )
    right  $\leftarrow$  SillySort( $A[\lfloor n/2 \rfloor + 1 \dots n]$ )
     $B \leftarrow \text{concatenate}(\text{left}, \text{right})$ 
    return MergeSort( $B$ )

```



Here, *MergeSort* is a function that sorts its input list using the merge sort algorithm.

Let $T(n)$ denote the worst-case running time of SillySort on an input of size n . Assume that concatenation of two lists of total size n takes $\Theta(n)$ time.

Which of the following recurrence relations correctly represents $T(n)$?

Your Answer

Not answered

Correct Answer

B. $T(n) = 2T(n/2) + \Theta(n \log n)$

Option A

$$T(n) = 2T(n/2) + \Theta(n)$$

Option B

CORRECT

$$T(n) = 2T(n/2) + \Theta(n \log n)$$

Option C

$$T(n) = T(n/2) + \Theta(n \log n)$$

Option D

$$T(n) = 2T(n/2) + \Theta(\log n)$$

OFFICIAL SOLUTION

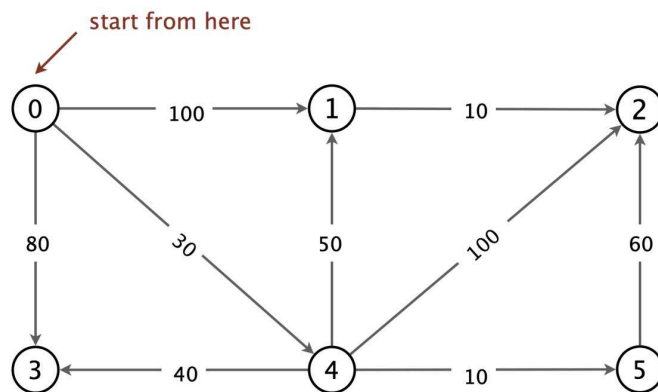
Answer: B

[Official discussion page](#)

Q13 | Technical

UNATTEMPTED

Consider the following edge-weighted directed graph **G**:



The Bellman–Ford algorithm is run on this graph with source vertex $s = 0$.

The value $\text{distTo}[v]$ denotes the *current shortest distance known from the source vertex s to vertex v , based on the edges relaxed so far*.

Within a single pass of the algorithm, the edges are relaxed strictly in the following order:

$0 \rightarrow 1, 0 \rightarrow 3, 0 \rightarrow 4, 1 \rightarrow 2, 4 \rightarrow 1, 4 \rightarrow 2, 4 \rightarrow 3, 4 \rightarrow 5, 5 \rightarrow 2$

Initially, $\text{distTo}[0] = 0$ and for all other vertices $\text{distTo}[v] = \infty$.

Immediately after the first pass of the Bellman–Ford algorithm, which of the following statements are correct?

Your Answer

Not answered

Correct Answer

A. $\text{distTo}[1] = 80$, C. $\text{distTo}[3] = 70$, D. $\text{distTo}[4] = 30$

Option A

$\text{distTo}[1] = 80$

CORRECT

Option B

$\text{distTo}[2] = 110$

Option C

CORRECT

distTo[3] = 70**Option D**

CORRECT

distTo[4] = 30**OFFICIAL SOLUTION****Answer:** A, C, D[↗ Official discussion page](#)**Q14 | Technical**

UNATTEMPTED

On a computer with a **32**-bit memory address, the length of the memory location of **1** byte, is installed set-associative cache. The cache size is **16** KB, the block (line) size is **16** bytes, and the set associative cache is **4**-way. Into which set (in decimal) is mapped the content of the memory address **10FFCFF**_(HEX) ?

Your Answer

Not answered

Correct Answer

207**Accepted Answer**

207

OFFICIAL SOLUTION**Answer:** Not available[↗ Official discussion page](#)**Q15 | Technical**

UNATTEMPTED



A sender uses Go-Back-N ARQ with a window size of **3**. A total of **10** data frames must be transmitted. The channel is such that every **5**th transmission attempt made by the sender is lost (the lost transmission is a data frame; ACKs are never lost). How many total data transmissions (original transmissions plus all required retransmissions) are needed for the sender to successfully deliver all **10** frames?

Your Answer

Not answered

Correct Answer

18**Accepted Answer**

18

OFFICIAL SOLUTION**Answer:** Not available[↗ Official discussion page](#)

Q16 | Technical

UNATTEMPTED

Sender **A** transmits packets to receiver B using the Go-Back-**N** ARQ protocol with a window size of **N = 3**. Packets are indexed (numbered) starting from **0**. During transmission, packet number **1** (**pkt1**) and packet number **4** (**pkt4**) are lost. After the second timeout occurs due to the loss of packet **4**, **A** must retransmit the required set of packets.

Which set of packets does **A** retransmit after the second timeout?

Your Answer

Not answered

Correct Answer

C. {pkt4, pkt5, pkt6}

Option A

{pkt4}

Option B

{pkt4, pkt5}

Option C

{pkt4, pkt5, pkt6}

CORRECT

Option D

{pkt5, pkt6}

OFFICIAL SOLUTION**Answer: C**[➤ Official discussion page](#)**Q17 | Technical**

UNATTEMPTED

A system has a total of **10** instances of a resource. The current allocation and maximum demand of each process is:

Process	Allocated	Maximum
P_1	1	6
P_2	1	5
P_3	2	4
P_4	4	7

If one additional request of the resource is made by a process, which of the following statements about the system's safety is/are correct?

Your Answer

Correct Answer

Not answered

B. If P_4 requests one more instance of the resource, the system becomes unsafe because not enough free resources remain. **D. If P_3 requests one more instance of the resource, the system stays safe and a valid safe sequence exists.**

Option A

If P_4 requests one more instance of the resource, the system stays safe.

Option B

CORRECT

If P_4 requests one more instance of the resource, the system becomes unsafe because not enough free resources remain.

Option C

If P_3 requests one more instance of the resource, the system becomes unsafe.

Option D

CORRECT

If P_3 requests one more instance of the resource, the system stays safe and a valid safe sequence exists.

OFFICIAL SOLUTION

Answer: B, D

[↗ Official discussion page](#)

Q18 | Technical

UNATTEMPTED

Consider two processes attempting to enter the critical section using the following protocol:

- Each process P_i sets **flag[i] = true** to declare intent.
- Then it waits while **flag[1 - i] = true**.

**Process P_0** **Process P_1**

```
flag[0] = true;
while (flag[1]==true);
```

```
/* Critical Section */
```

```
flag[0] = false;
```

```
flag[1] = true;
while (flag[0]==true);
```

```
/* Critical Section */
```

```
flag[1] = false;
```

Both **flag[0]** and **flag[1]** are initially **false**.

Which of the following statements is/are TRUE?

Your Answer

Not answered

Correct Answer

A. It guarantees mutual exclusion. **C. It may cause both threads to wait forever, violating progress.** **D. It guarantees bounded waiting.**

Option A

CORRECT

It guarantees mutual exclusion.

Option B

It guarantees progress but violates mutual exclusion.

Option C

CORRECT

It may cause both threads to wait forever, violating progress.

Option D

CORRECT

It guarantees bounded waiting.

OFFICIAL SOLUTION**Answer:** A, C, D[↗ Official discussion page](#)**Q19 | Technical**

UNATTEMPTED

A queue Q (front rear) initially contains $\text{\textcolor{red}{verb}}|[1, 2, 3, 4]|$.A stack S is empty.

You may use only the following operations:

- $\text{\textcolor{red}{verb}}|Enqueue(x)|$ - insert into rear of Q
- $\text{\textcolor{red}{verb}}|Dequeue()|$ - remove from front of Q
- $\text{\textcolor{red}{verb}}|Push(x)|$ - push into S
- $\text{\textcolor{red}{verb}}|Pop()|$ - pop from S

You must transfer all elements from Q to S such that:

- The front element of Q becomes the top of S , and
- The relative order of all elements is preserved.

What is the minimum number of $\text{\textcolor{red}{verb}}|Push|$ operations required?

Your Answer

Not answered

Correct Answer

4**Accepted Answer**

4

OFFICIAL SOLUTION

Answer: 4[➤ Official discussion page](#)**Q20 | Technical**

UNATTEMPTED

Consider inserting the integer keys $A = [67, 13, 49, 24, 40, 33, 58]$ into a hash table of size **9** using the hash function $h(k) = (11k + 4) \bmod 9$. Collisions are resolved using **chaining**, where new keys are appended at the **end** of the chain.

After all keys are inserted in the given order, which of the following is/are correctly describing the final state of the hash table?

Your Answer

Not answered

Correct Answer

A. Bucket 3 contains the chain **$67 \rightarrow 13 \rightarrow 49 \rightarrow 40 \rightarrow 58$ and bucket 7 contains $24 \rightarrow 33$** **Option A**

CORRECT

Bucket 3 contains the chain **$67 \rightarrow 13 \rightarrow 49 \rightarrow 40 \rightarrow 58$** and bucket 7 contains **$24 \rightarrow 33$** **Option B**Bucket 3 contains the chain **$67 \rightarrow 13 \rightarrow 49 \rightarrow 40 \rightarrow 58$** and bucket 6 contains **$24 \rightarrow 33$** **Option C**Bucket 4 contains the chain **$67 \rightarrow 49 \rightarrow 40$** and bucket 7 contains **$13 \rightarrow 24 \rightarrow 33 \rightarrow 58$** **Option D**Bucket 4 contains **$67 \rightarrow 49 \rightarrow 58$** and bucket 7 contains **$24 \rightarrow 33$** **OFFICIAL SOLUTION****Answer: A**[➤ Official discussion page](#)**Q21 | Technical**

UNATTEMPTED

What is the size of the largest max-binary-heap that is also a valid binary search tree? (Assume the heap has no duplicate elements)

Your Answer

Not answered

Correct Answer

2**Accepted Answer****2**

OFFICIAL SOLUTION**Answer: 2**[↗ Official discussion page](#)**Q22 | Technical**

UNATTEMPTED

Consider the following C program:

```
int* alpha(int *l, int *r) {  
    *l += *r;  
    return r;  
}  
  
int main() {  
    int x = 5, y = 7;  
    int *z = alpha(&x, &y);  
    *z += 3;  
    printf("%d %d", x, y);  
    return 0;  
}
```

What will be the output of the program?

Your Answer

Not answered

Correct Answer

B. 12 10**Option A**

12 7

**Option B**

12 10

CORRECT

Option C

5 10

Option D

5 7

OFFICIAL SOLUTION**Answer: B**[↗ Official discussion page](#)**Q23 | Technical**

UNATTEMPTED

Consider the following C program.

```
#include <stdio.h>
int main () {
    int a[3][3] = {
        {3, 5, 7},
        {2, 4, 6},
        {8, 9, 10}
    };

    printf("%d", *(*a + (a[0][0] == 3)));

    return 0;
}
```

The output of the program is _____.

Your Answer

Not answered

Correct Answer

5

Accepted Answer

5

OFFICIAL SOLUTION

Answer: 5

[↗ Official discussion page](#)

Q24 | Technical

UNATTEMPTED



Consider the following ANSI C program.

```
#include <stdio.h>
int main()
{
    int x = 0, y = 5;
    if ((x != 0) && (y++))
        x = y;
    printf("%d %d", x, y);
    return 0;
}
```

Which one of the following options is correct?

Your Answer

Not answered

Correct Answer

**A. 0 5
**

Option A

0 5

CORRECT

Option B

5 6

Option C

6 6

Option D

The program will not compile.

OFFICIAL SOLUTION**Answer: A**[↗ Official discussion page](#)**Q25 | Technical**

UNATTEMPTED

Let $\mathbf{x}$ and $\mathbf{y} \in \mathbb{R}^n$ be non-zero column vectors, form the matrix $\mathbf{A} = \mathbf{xy}^T$, where $\mathbf{y}^T$ is the transpose of $\mathbf{y}$. Then the rank of $\mathbf{A}$ is:

Your Answer

Not answered

Correct Answer

D. None of the above**Option A**

2

Option B

0

Option CAt least $n/2$ **Option D**

None of the above

CORRECT

OFFICIAL SOLUTION**Answer: D**[↗ Official discussion page](#)**Q26 | Technical**

UNATTEMPTED

In a certain computer, one instruction is executed in the order of steps **1** through **6** in the table shown below. How many nanoseconds are required to execute **6** instructions using pipeline processing in the figure shown below? Here, it takes **10** nanoseconds to execute each step, and there is no instruction, such as branch and jump, that stalls the pipeline processing.

Table Execution Steps Per Instruction

Step	
1	Fetch instruction code
2	Decode instruction
3	Fetch address
4	Calculate effective address
5	Fetch data
6	Execute operation

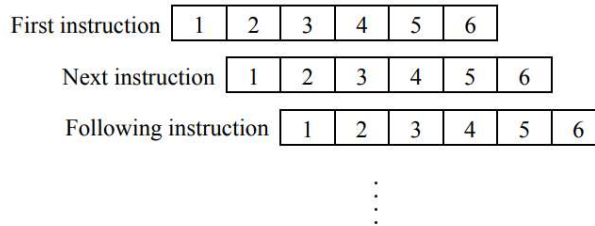


Fig. Pipeline Processing to Execute Instructions

Your Answer

Not answered

Correct Answer

110

Accepted Answer

110

OFFICIAL SOLUTION

Answer: Not available

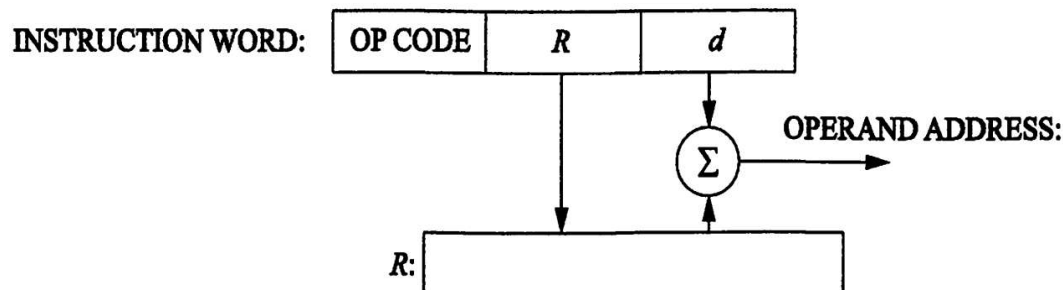
[Official discussion page](#)

Q27 | Technical

UNATTEMPTED



When a CPU fetches an instruction word from memory, the word contains an operation code (op code) that indicates the type of operation the CPU is to perform and information specifying where the instruction operands are located. A computer may use various addressing modes to specify the operand location. One such addressing mode is shown below, where R designates some register within the CPU and d is a constant embedded in the instruction word.



Which of the following terms best describes the addressing mode used by the instruction above ? (Σ indicates addition)

Your Answer

Correct Answer

Not answered

C. Indexed addressing

Option A

Immediate addressing

Option B

Direct addressing

Option C

Indexed addressing

CORRECT

Option D

Indirect addressing

OFFICIAL SOLUTION**Answer: C**[↗ Official discussion page](#)**Q28 | Technical**

UNATTEMPTED

When data is transmitted in a serial format, an error detection bit called a parity bit may be appended to the bit stream. A serial receiver that uses even parity has received the following nine bits (parity bit followed by eight data bits).

000110100

Which statement best describes the received code?

Your Answer

Not answered

Correct Answer

C. An odd number of bits have been corrupted during transmission.**Option A**

No errors occurred during transmission.

Option B

An even number of bits have been corrupted during transmission.

Option C

An odd number of bits have been corrupted during transmission.

CORRECT

Option D

A bit transmitted as a one (1) has been changed to a zero (0) during transmission.

OFFICIAL SOLUTION**Answer: C**[↗ Official discussion page](#)**Q29 | Technical****UNATTEMPTED**

Which of the following statements is/are true?

Your Answer

Not answered

Correct Answer

A. The union of two context-free languages is context-free.**Option A****CORRECT**

The union of two context-free languages is context-free.

Option B

The union of two context-free languages is not context-free.

Option C

The intersection of two context-free languages is context-free.

Option D

The intersection of two context-free languages is not context-free.

OFFICIAL SOLUTION**Answer: A**[↗ Official discussion page](#)**Q30 | Technical****UNATTEMPTED**Let $A = \{2, 4, 6, 8, 10\}$ and $B = \{3, 5, 7, 8, 9, 11, 13\}$. Find the total number of one-one maps $f : A \rightarrow B$ such that $f(2) + f(4) = 16$.

Your Answer

Not answered

Correct Answer

360**Accepted Answer**

360

OFFICIAL SOLUTION**Answer: Not available**

[➤ Official discussion page](#)
Q31 | Technical

UNATTEMPTED

A sequential logic circuit has one input (x), one output (z), and six states labeled **A** – **F**. The circuit is described by the state table shown below. Entries in the two right-hand columns represent the next-state/output-value combination for each present-state condition and input value. For example, if the present state is **B** and an input $x = 1$ is applied, the next state will be **D** and the value of (z) will be **0**.

If the circuit is initially in State **C** and the input sequence $x = 100$ is applied (meaning the first input is **1**, the second input is **0**, etc.), the output sequence is best described by

Present State	Next State/Output Input x	
	0	1
A	C/0	A/0
B	B/1	D/0
C	D/0	B/1
D	F/1	B/1
E	E/1	F/0
F	F/0	A/1

Your Answer

Not answered

Correct Answer

D. 111

Option A

101

Option B

010

Option C

100

Option D

111

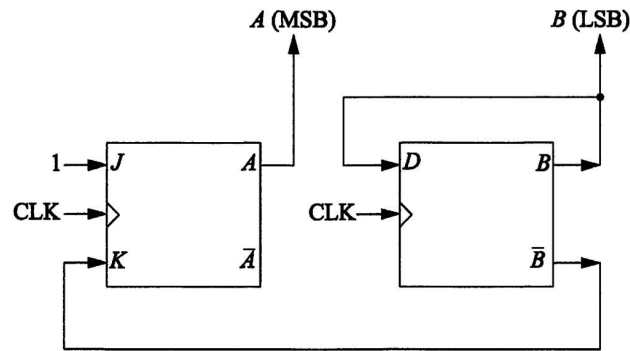
CORRECT

OFFICIAL SOLUTION**Answer: D**
[➤ Official discussion page](#)

Q32 | Technical

UNATTEMPTED

Flip-flops A and B form a sequential synchronous circuit as shown below.



After the clock pulse, binary count $10(A = 1, B = 0)$ changes to:

Your Answer

Not answered

Correct Answer

A. 00

Option A

00

CORRECT

Option B

01

Option C

10

Option D

11

OFFICIAL SOLUTION

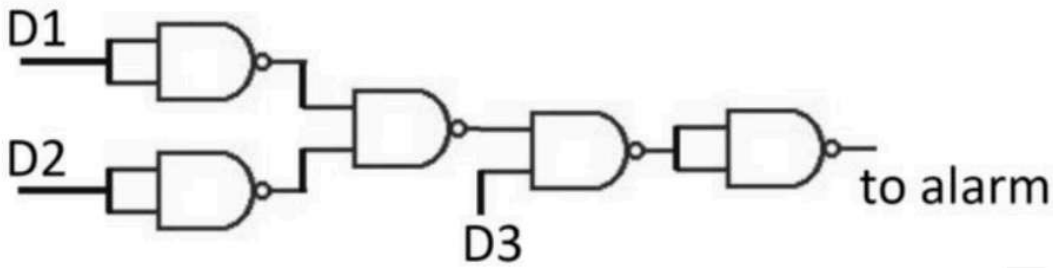
Answer: A

[Official discussion page](#)


Q33 | Technical

UNATTEMPTED

Three nearby laboratory rooms **1, 2, 3** have one smoke detector each labelled **D1, D2, D3** respectively, for fire safety. In case smoke is detected, the detector output goes to a logic state *True*. However, to prevent false alarm, these outputs are connected to the logic circuit below whose output activates a single alarm.



In which of the following cases will the alarm definitely ring:

Your Answer

Not answered

Correct Answer

B. Smoke is detected in rooms 2 and 3

Option A

Smoke is detected in room **3**

Option B

Smoke is detected in rooms **2** and **3**

CORRECT

Option C

Smoke is detected in rooms **1** or **3**

Option D

Smoke is detected in any two of three rooms

OFFICIAL SOLUTION

Answer: B

[Official discussion page](#)



Q34 | Technical

UNATTEMPTED

Indexes are auxiliary access structures used to increase the speed of record retrieval in response to certain search conditions.

Based on knowledge about indexes, which of the following statements is/are correct?

Your Answer

Not answered

Correct Answer

**B. An index file is an example of a sequential file; the key-pointer pairs can be treated as records ordered by the value of the search key.
, D. Inserting or deleting records in the data file results in the same action on its index file (if it is dense),**

since a key-pointer pair for that record is inserted or deleted.

Option A

A **sparse index** has an index entry for each value of the search key (therefore, for each record) in the data file. A **dense index** has index entries for only some of the search key values.

Option B

CORRECT

An index file is an example of a **sequential file**; the key-pointer pairs can be treated as records ordered by the value of the search key.

Option C

A file can have, at most, **one secondary index**, used to physically order the records of the file on disk; however, a file may have **several primary indexes**, which can be specified on any field of a file.

Option D

CORRECT

Inserting or deleting records in the data file results in the same action on its index file (if it is **dense**), since a key-pointer pair for that record is inserted or deleted.

OFFICIAL SOLUTION

Answer: B, D

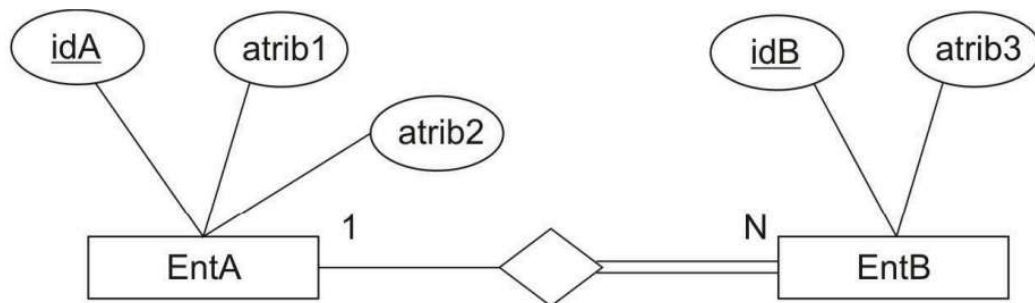
[Official discussion page](#)

Q35 | Technical

UNATTEMPTED



Analyze the following **Entity-Relationship (ER) diagram**.



Consider the ER diagram in which an entity of type **EntA** can be related to several entities of type **EntB**, and each entity of type **EntB** is related to one entity of type **EntA**.

If this diagram is converted to the **relational model**, which of the following sets of tables presents the **best mapping that satisfies Third Normal Form (3NF)**?

Your Answer

Not answered

Correct Answer

C. EntA (idA, atrib1, atrib2), EntB(idB, atrib3, idA).

Option A

EntA (idA, atrib1, atrib2), EntB (idB, atrib3).

Option B

EntAB (idA, idB, atrib1, atrib2, atrib3).

Option C**CORRECT**

EntA (idA, atrib1, atrib2), EntB(idB, atrib3, idA).

Option D

EntA (idA, atrib1, atrib2), AB (idA, idB), EntB (idB, atrib3).

OFFICIAL SOLUTION**Answer: C**
[↗ Official discussion page](#)
Q36 | Technical**UNATTEMPTED**

Let $G = (V, E)$ be a simple, connected, **non-Eulerian** graph. We want to construct a graph H (not necessarily simple) that is **Eulerian** and that contains G as a subgraph. Consider the following possible construction processes:

- (I) Add a new vertex, connecting it to every vertex of G by an edge.
- (II) Add a new vertex, connecting it to every vertex of **odd degree** in G by an edge.
- (III) Create a new copy G' of graph G and add an edge connecting each pair of corresponding vertices.
- (IV) Duplicate all edges of G .
- (V) Add edges to G until the **complete graph** with $|V|$ vertices is formed.

Which of the above processes **always** correctly constructs the graph H ?

- A. All
- B. Only (II), (III), (IV) and (V)
- C. Only (II), (III), and (IV)
- D. Only (II) and (IV)

Your Answer

Not answered

Correct Answer

D

Accepted Answer

D

OFFICIAL SOLUTION**Answer:** D[➤ Official discussion page](#)**Q37 | Technical**

UNATTEMPTED

Consider the following context-free grammar G .

$$S \rightarrow bABaa \mid Sa \mid a$$

$$A \rightarrow aB$$

$$B \rightarrow baB \mid \epsilon$$

Let L_A and L_B be the languages consisting of the terminal strings that can be derived from the variables A and B , respectively.

Which of the following options is/are correct?

Your Answer

Not answered

Correct Answer

A. L_A is regular.
B. L_B is regular.
C. $L(G)$ is regular.

Option A

CORRECT

 L_A is regular.**Option B**

CORRECT

 L_B is regular.**Option C**

CORRECT

 $L(G)$ is regular.**Option D** $L(G)$ is a context-free language but not regular.**OFFICIAL SOLUTION****Answer:** A, B, C[➤ Official discussion page](#)**Q38 | Technical**

UNATTEMPTED

Given items $x_1, x_2, \dots, x_n$, where each item x_i has a non-negative weight $w(x_i)$. You are required to select a subset of items such that no two selected items are adjacent (i.e., if x_i is selected, then x_{i-1} and x_{i+1} cannot be selected).

Let W_i denote the maximum total weight that can be obtained by selecting from items x_1 through x_i .

For the purpose of this question, assume that the recurrence is applied for $i \geq 2$. You do not need to worry about the base cases.

Which of the following recurrence relations correctly computes W_{i+1} ?

Your Answer

Not answered

Correct Answer

C. $W_{i+1} = \max\{W_i, W_{i-1} + w(x_{i+1})\}$

Option A

$W_{i+1} = \max\{W_i, W_i + w(x_{i+1})\}$

Option B

$W_{i+1} = \max\{W_{i-1}, W_i + w(x_{i+1})\}$

Option C

$W_{i+1} = \max\{W_i, W_{i-1} + w(x_{i+1})\}$

CORRECT

Option D

$W_{i+1} = W_{i-1} + w(x_{i+1})$

OFFICIAL SOLUTION

Answer: C

[Official discussion page](#)



Q39 | Technical

UNATTEMPTED

You are given a black box implementation of a comparison-based data structure that supports the operations insert and extract-min. It is known that the data structure can be constructed on n elements in time $\Theta(n \log \log n)$.

A sorting algorithm is implemented using this black box as follows:

- Build the data structure on the n elements.
- Repeatedly perform extract-min until all elements are removed.

If each extract-min operation takes $T(n)$ time, which of the following must be true?

Your Answer

Not answered

Correct Answer

A. $T(n) = \Omega(\log n)$

Option A

CORRECT

$$T(n) = \Omega(\log n)$$

Option B

$$T(n) = \Omega(\log \log n)$$

Option C

$$T(n) = \Omega(1)$$

Option D

$$T(n) = \Omega(\sqrt{\log \log n})$$

OFFICIAL SOLUTION**Answer: A**
[↗ Official discussion page](#)
Q40 | Technical

UNATTEMPTED

Assume two end-hosts communicate using the sliding window protocol. The size of the sender window is w bits. Let C be the link bandwidth between the two end-hosts in bps. Further, the round trip time (representing here the time when the first bit of the segment is transmitted till its acknowledgment is received) on the path between the two end-hosts is RTT seconds.

If the roundtrip time satisfies $RTT > \frac{w}{C}$, what is the **maximum throughput** achieved by the two end-hosts?

Your Answer

Not answered

Correct Answer

B. $\frac{w}{RTT}$ **Option A**

$$C$$

Option B

CORRECT

$$\frac{w}{RTT}$$

Option C

$$\frac{C}{RTT}$$

Option D

$$\frac{w}{C \cdot RTT}$$

OFFICIAL SOLUTION**Answer: B**

[➤ Official discussion page](#)**Q41 | Technical**

UNATTEMPTED

A file of size F bits is sent from Host A to Host B using **store-and-forward** packet switching across 3 hops ($A \rightarrow S_1 \rightarrow S_2 \rightarrow B$). The file is divided into segments of S bits, and an **80-bit header** is added to each segment. Thus, each packet size is **$S + 80$** bits. All links have transmission rate R bps. Propagation delays are ignored. Which of the following expressions gives the total time to deliver the entire file to Host B?

Your Answer

Not answered

Correct Answer

c. $\frac{S + 80}{R} \left(\frac{F}{S} + 2 \right)$

Option A

$$\frac{S + 80}{R} \left(\frac{F}{S} \right)$$

Option B

$$\frac{S + 80}{R} \left(\frac{F}{S} + 1 \right)$$

Option C

$$\frac{S + 80}{R} \left(\frac{F}{S} + 2 \right)$$

CORRECT

Option D

$$\frac{F}{R} + 2 \frac{S + 80}{R}$$

**OFFICIAL SOLUTION****Answer: C**[➤ Official discussion page](#)**Q42 | Technical**

UNATTEMPTED

A TCP sender wants to transmit a **402 KB** file to a receiver over an already established TCP connection. At time t , the sender's congestion window is **$cwnd = 2 \text{ KB}$** , the slow-start threshold is **$ssthresh = 64 \text{ KB}$** , and the sender is currently in the slow-start phase. The Maximum Segment Size (MSS) is **2 KB**, the receiver advertises a window of **70 KB**, ACKs are never lost, and the sender always has data available to send. Assuming standard TCP congestion control, how many RTTs are required for the sender to successfully receive the ACK for the last byte of the **402 KB** file?

Your Answer

Correct Answer

Not answered

C. 11 RTTs

Option A

9 RTTs

Option B

10 RTTs

Option C

11 RTTs

CORRECT

Option D

12 RTTs

OFFICIAL SOLUTION**Answer: C**[↗ Official discussion page](#)**Q43 | Technical**

UNATTEMPTED

A system with a single CPU uses priority-based preemptive scheduling with three processes(X, Y, and Z):

X = highest priority, Y = medium priority, Z = lowest priority.

Processes can be in one of five states:

RUNNING, READY, BLOCKED, TERMINATED, Not Yet Created.

The steps below describe all activity of the system.

There are no extra CPU bursts, no extra I/O events beyond what is written here.

Your Answer

Not answered

Correct Answer

A. Step 1: Process X is loaded into memory and X starts running.

Option A

Step 1: Process X is loaded into memory and X starts running.

CORRECT

Option BStep 2: X calls `fork()` → Y is created.**Option C**

Step 3: The running process performs an I/O request to the disk.

Option D

Step 4: The running process calls `fork()` → Z is created.

Option E

Step 5: The previously issued I/O request completes.

Option F

Step 6: The running process finishes execution.

OFFICIAL SOLUTION

Answer: A

[↗ Official discussion page](#)

Q44 | Technical

UNATTEMPTED

A system uses two-level paging, where both the outer page table and the inner page tables use the same number of index bits.

The page size is **32** bytes, the virtual address size is **15** bits, and physical memory consists of **128** pages.

The Page Table Base Register (PTBR) contains physical page number **51**. Each Page Table Entry (PTE) is one byte long, where the most significant bit is the valid bit and the remaining seven bits represent the Physical Frame Number (PFN).

Translate the following virtual address using the physical page contents shown below:

$$VA = 0x5C5B$$

Page 51		Page 86		Page 103	
Index	Byte	Index	Byte	Index	Byte
0	cd	0	10	0	07
1	a6	1	00	1	09
2	d2	2	e7	2	17
⋮	⋮	⋮	⋮	⋮	⋮
21	82	24	16	26	15
22	d7	25	1c	27	1d
23	d6	26	11	28	05
24	c9	27	1a	29	14
⋮	⋮	⋮	⋮	⋮	⋮
31	a7	31	19	31	14



What is the final physical address corresponding to the given virtual address?

Your Answer

Not answered

Correct Answer

C. 0xCFB

Option A

0x6FB

Option B

0x001D

Option C

0xCFB

CORRECT

Option D

None of the above

OFFICIAL SOLUTION**Answer: C**[↗ Official discussion page](#)**Q45 | Technical**

UNATTEMPTED

Consider two threads executing concurrently. The semaphores s and m are counting semaphores shared between the two threads.

Both threads execute inside infinite loops, and their executions may interleave arbitrarily.

The code executed by the two threads is shown below:

Thread 1	Thread 2
<pre>while (true) { P(s); P(s); P(m); printf("a"); V(m); }</pre>	<pre>while (true) { P(m); printf("b"); V(m); V(s); }</pre>



Which of the following statements is/are **TRUE**?

Your Answer

Not answered

Correct Answer

C. If initially $s = 2$ and $m = 1$, it is possible for two consecutive `<code>"b"</code>` characters to appear in the output.

Option A

If initially $s = 0$ and $m = 1$, then every printed `<code>"a"</code>` must appear immediately after two printed `<code>"b"</code>` characters.

Option B

If initially $s = 0$ and $m = 1$, it is possible for `"a"` to appear before any `"b"`.

Option C

CORRECT

If initially $s = 2$ and $m = 1$, it is possible for two consecutive `"b"` characters to appear in the output.

Option D

If initially $s = 2$ and $m = 1$, every printed `"a"` must appear after at least one printed `"b"`.

OFFICIAL SOLUTION

Answer: C

[Official discussion page](#)

Q46 | Technical

UNATTEMPTED

A perfect binary tree with exactly 7 distinct keys is constructed. The keys are visited using an in-order traversal, producing the sequence:

8, 3, 7, 17, 1, -1, 12

Which of the following statements is the pre-order traversal of the tree?

A. 17, 3, 8, 7, -1, 1, 12

B. 17, 8, 3, 7, -1, 1, 12

C. 3, 8, 7, 17, -1, 1, 12

D. The pre-order traversal of the tree cannot be uniquely determined from the given information.



Your Answer

Not answered

Correct Answer

A. Option A

Option A

CORRECT

Option A

Option B

Option B

Option C

Option C

Option D

Option D

OFFICIAL SOLUTION

Answer: A[↗ Official discussion page](#)**Q47 | Technical**

UNATTEMPTED

A singly linked list of integers is represented using the following structure. Consider the recursive function shown below.

```
struct node {  
    int data;  
    struct node* next;  
};  
  
typedef struct node Node;  
  
void fun_list(Node* head)  
{  
    if (head == NULL) {  
        return;  
    }  
  
    printf("%d ", head->data);  
  
    if (head->next != NULL)  
        fun_list(head->next);  
  
    printf("%d ", head->data);  
}
```

The function `fun_list` is called with the head pointer of the linked list:

1 → 2 → 3 → 4 → 5

What will be printed on the output?

Your Answer

Not answered

Correct Answer

C. 1 2 3 4 5 5 4 3 2 1**Option A**

1 2 3 4 5

Option B

1 2 3 4 5 4 3 2

Option C

1 2 3 4 5 5 4 3 2 1

CORRECT

Option D

5 4 3 2 1 1 2 3 4 5

OFFICIAL SOLUTION**Answer: C**

[➤ Official discussion page](#)**Q48 | Technical**

UNATTEMPTED

Which of the following code snippets correctly creates a 3×5 two-dimensional array on the heap using dynamic memory allocation in C?

(Here, 3×5 means **3 rows** and **5 columns**.)

Your Answer

Not answered

Correct Answer

A. `int **a = malloc(sizeof(int *) * 3); for (int i = 0; i < 3; i++) { a[i] = malloc(sizeof(int) * 5); }`

Option A

CORRECT

```
int **a = malloc(sizeof(int *) * 3);
for (int i = 0; i < 3; i++) {
    a[i] = malloc(sizeof(int) * 5);
}
```

Option B

```
int **a = malloc(sizeof(int *) * 5);
for (int i = 0; i < 5; i++) {
    a[i] = malloc(sizeof(int) * 3);
}
```

Option C

```
int *a = malloc(sizeof(int) * 15);
```

Option D

```
int *a = malloc(sizeof(int) * 3);
for (int i = 0; i < 3; i++) {
    a[i] = malloc(sizeof(int) * 5);
}
```

OFFICIAL SOLUTION**Answer: A**[➤ Official discussion page](#)**Q49 | Technical**

UNATTEMPTED

In a computer with cache, we have the average number of clock periods per instruction equal to **4**, if there are no misses in the cache. What is the real number of clock periods per instruction, if the probability of miss in the cache is **10%**? For the replacement of the block (line) in the cache, we need **5** clock periods for read and **10** for write accesses. Assume that each instruction requires an average of **2** memory accesses and that **20%** of all are write accesses. (Write answer up to one decimal place)

Your Answer

Not answered

Correct Answer

5.2

Accepted Answer

5.2

OFFICIAL SOLUTION**Answer:** 5, 2[➤ Official discussion page](#)**Q50 | Technical**

UNATTEMPTED

Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$, $C = \begin{bmatrix} 4 & 5 & 6 \\ 1 & 2 & 3 \\ 7 & 8 & 9 \end{bmatrix}$ and $D = \begin{bmatrix} -1 & 2 & 3 \\ 4 & -5 & 6 \\ 7 & 8 & -9 \end{bmatrix}$.

Which of the following statements are **true**?

Your Answer

Not answered

Correct Answer

A. $|A| = |B|$, **C.** $|B| = -|C|$ **Option A**

$|A| = |B|$

CORRECT

Option B

$|C| = |D|$

Option C

$|B| = -|C|$

CORRECT

Option D

$|A| = -|D|$

OFFICIAL SOLUTION**Answer:** A, C[➤ Official discussion page](#)

Q51 | Technical

UNATTEMPTED

Suppose A and B are independent boolean variables. Determine the missing entries x and y in the joint distribution of $P(A, B)$ shown below:

Given:

- $P(A = T, B = T) = 0.25$
- $P(A = T, B = F) = 0.35$
- $P(A = F, B = T) = x$
- $P(A = F, B = F) = y$

Your Answer

Not answered

Correct Answer

A. $x = \frac{1}{6}, y = \frac{7}{30}$

Option A

$$x = \frac{1}{6}, y = \frac{7}{30}$$

CORRECT

Option B

$$x = \frac{1}{10}, y = \frac{3}{10}$$

Option C

$$x = 0.2, y = 0.2$$

Option D

$$x = \frac{1}{5}, y = \frac{1}{5}$$

OFFICIAL SOLUTION

Answer: A

[Official discussion page](#)



Q52 | Technical

UNATTEMPTED

Consider the attribute set $R = ABCDEF$ and the functional dependency set

$$F = \{C \rightarrow D, A \rightarrow B, B \rightarrow EF, F \rightarrow A\}$$

A BIJZLI Classes student decides to write a decomposition of tables as $ABDE$ and $ABCF$. Right now, this is

not a lossless decomposition. Which of the following changes (applied individually, not combined together) would make this a lossless decomposition?

- A. Adding $D \rightarrow ABCDEF$ to the functional dependency set F
- B. Adding $E \rightarrow C$ to the functional dependency set F
- C. Adding $B \rightarrow D$ to the functional dependency set F

Your Answer

Not answered

Correct Answer

D. Only B and C

Option A

All A, B, C

Option B

Only B

Option C

Only C

Option D

Only B and C

CORRECT

OFFICIAL SOLUTION

Answer: D

[Official discussion page](#)



Q53 | Technical

UNATTEMPTED

Let T_1 and T_2 be two transactions being processed by a DBMS. The terms **lockR** and **lockW** correspond to requests for read and write locks, respectively, and **Unlock** corresponds to releasing a lock. A, B , and C are database data items.

The excerpt below is a fragment of the schedule of T_1 and T_2 defined by the DBMS scheduler (the fragment is not complete):

```
start(T1); lockR(T1, A); read (T1, A); start(T2);
lockR(T2, B); read (T2, B); lockW (T1, C); read(T1,C);
write(T1,C); unlock(T1, C); lockW (T1, B); lockW (T2, A); lockR(T2,C);
...
```

Consider the following statements:

- I. The fragment shows an example of the application of the **2PL protocol** (two-phase locking).
- II. The fragment violates the **2PL protocol**.
- III. The fragment shows an example in which there is a **deadlock** between **T1** and **T2**.
- IV. The fragment does **not** have a deadlock between **T1** and **T2**.
- V. Nothing can be asserted.

The correct statements are:

Your Answer

Not answered

Correct Answer

**C. Only II and III
**

Option A

Only I and III

Option B

Only II and IV

Option C

Only II and III

CORRECT

Option D

Only I and IV



OFFICIAL SOLUTION

Answer: C

[➤ Official discussion page](#)

Q54 | Technical

UNATTEMPTED

Let a relation **R** have four tuples — **(12, 15, 5100)**, **(13, NULL, 3500)**, **(14, NULL, NULL)**, and **(15, 12, NULL)**— in which the first, second, and third values in each tuple refer to attributes **at1**, **at2**, and **at3**, respectively. The following commands represent queries on **R** :

```
(C1) select * from R
      where (at1>=12) AND (at2>14)
(C2) select * from R
      where (at2>12) OR (at3>3000)
```

(C3) select * from R
where (NOT (at1<at2))

The number of tuples returned by the execution of commands (C1), (C2), and (C3), respectively, is:

Your Answer

Not answered

Correct Answer

D. one, two, and one.

Option A

two, one, and two.

Option B

two, two, and one.

Option C

one, one, and two.

Option D

one, two, and one.

CORRECT

OFFICIAL SOLUTION

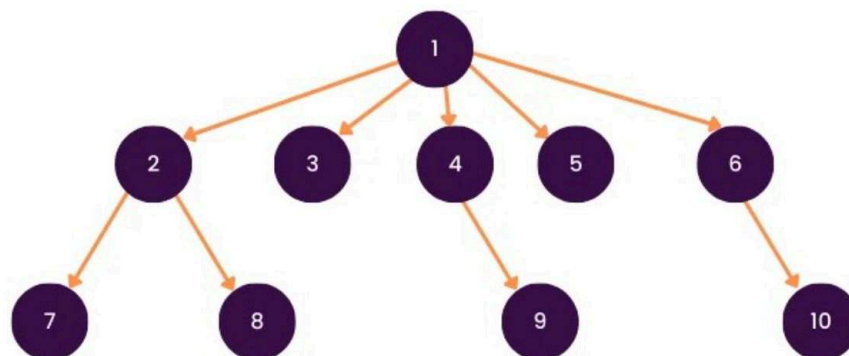
Answer: D

[➤ Official discussion page](#)

Q55 | Technical

UNATTEMPTED

Suppose we have a relation $\mathbf{T(P, C, I)}$ representing the following tree. That is, nodes are represented by distinct integers, and the meaning of a tuple (p, c, i) in $\mathbf{T}$ is that node c is the i th child of node p , from the left. Attribute $\mathbf{C}$ is the primary key in $\mathbf{T}$. For every node, except the root, we have a tuple in $\mathbf{T}$.



The following relational algebra query is made on the database.

$$R1(C1, C2) := \pi_{C1, C2} (\sigma_{I2 > I1} (\rho_{T1(P, C1, I1)}(T) \bowtie \rho_{T2(P, C2, I2)}(T)))$$

$$R2(C1, C3) := \pi_{C1, C3} (\sigma_{I2 > I1 \text{ AND } I3 > I2} (\rho_{T1(P, C1, I1)}(T) \bowtie \rho_{T2(P, C2, I2)}(T) \bowtie \rho_{T3(P, C3, I3)}(T)))$$

$$ANSWER := R1 - R2$$

The number of rows in ANSWER is?

Your Answer

Not answered

Correct Answer

5

Accepted Answer

5

OFFICIAL SOLUTION

Answer: 5

[Official discussion page](#)

Q56 | Technical

UNATTEMPTED

Consider the following program P_0 written in a hypothetical C-like language:

Program P_0

```
while (get() == 0);
halt;
```

Your Answer

Not answered

Correct Answer

B. Each call to `<code>get()</code>` asks the environment (e.g., a human user) to input 0 or 1 and returns the response.



Option A

Every value in this language is either 0 or 1.

Option B

CORRECT

Each call to `<code>get()</code>` asks the environment (e.g., a human user) to input 0 or 1 and returns the response.

Option C

The `<code>halt</code>` statement ends the program execution immediately.

Option D

After reaching a `<code>halt</code>` instruction, no further input is possible.

OFFICIAL SOLUTION**Answer:** B[➤ Official discussion page](#)**Q57 | Technical**

UNATTEMPTED

For any two sets S and T , $S \Delta T$ is defined as $(S \cup T) \setminus (S \cap T)$. Let A , B and C be sets such that $A \cap B \cap C = \phi$, and the number of elements in each of $A \Delta B$, $B \Delta C$ and $C \Delta A$ equals 100. Then the number of elements in $A \cup B \cup C$ equals

Your Answer

Not answered

Correct Answer

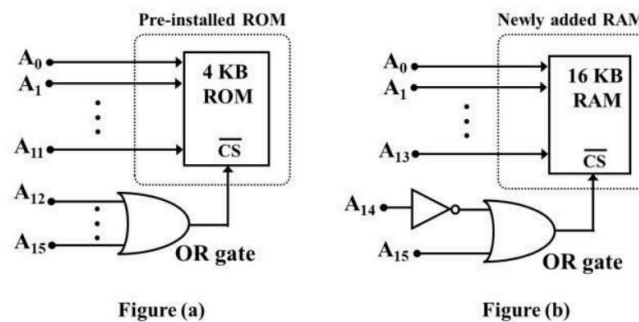
150**Accepted Answer**

150

OFFICIAL SOLUTION**Answer:** Not available[➤ Official discussion page](#)**Q58 | Technical**

UNATTEMPTED

An 8-bit microprocessor has 16-bit address bus ($A_{15} - A_0$) where A_0 is the LSB. As shown in Figure (a), it has a pre-installed 4 KB ROM whose starting address is 0000 H. The processor needs to be upgraded by adding a 16 KB RAM as shown in Figure (b). The address range for the newly added RAM is



Your Answer

Not answered

Correct Answer

C. 4000 H – 7FFF H**Option A****1000 H – 4FFF H****Option B****3000 H – 6FFF H**

Option C

4000 H – 7FFF H

CORRECT

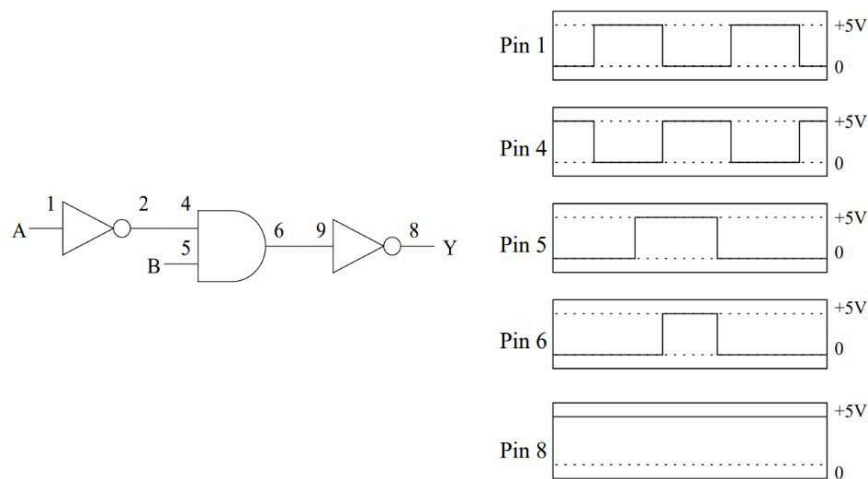
Option D

8000 H – BFFF H

OFFICIAL SOLUTION**Answer: C**[Official discussion page](#)**Q59 | Technical**

UNATTEMPTED

The digital electronic circuit shown below (left side) has some problem and is not performing as intended. The voltage at each pin as a function of time is shown in the adjacent figures.



The problem in the about circuit may be that

Your Answer

Not answered

Correct Answer

C. the Pin 8 is clamped to +5V**Option A**the **Pin 6** is shorted to ground**Option B**

the input inverter is shorted

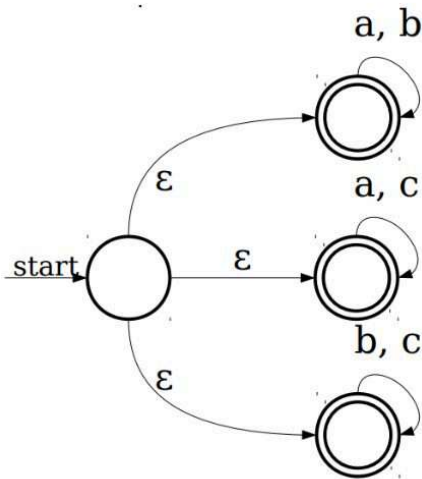
Option Cthe **Pin 8** is clamped to +5V

CORRECT

Option D**OR** gate is used instead of **AND** gate

OFFICIAL SOLUTION**Answer: C**[➤ Official discussion page](#)**Q60 | Technical****UNATTEMPTED**

Consider the following non-deterministic finite automaton.



Let p be the probability that a uniformly random string from $\{a, b, c\}^5$ is accepted by the above automaton. Which of the following is true about p ?

Your Answer

Not answered

Correct Answer

C. $p = 93/243$ **Option A** **$p = 32/243$** **Option B** **$p = 90/243$** **Option C** **$p = 93/243$** **CORRECT****Option D** **$p = 96/243$** **OFFICIAL SOLUTION****Answer: C**[➤ Official discussion page](#)

Q61 | Technical

UNATTEMPTED

Consider the following the LL(1) grammar:

$$S \rightarrow A$$

$$A \rightarrow xAC \mid yB$$

$$B \rightarrow yBC \mid \epsilon$$

$$C \rightarrow z$$

Assume we want to add one more production for C (i.e., of the form $C \rightarrow \alpha$) so that the resulting grammar becomes not LL(1). Adding which of the following productions would make the grammar NOT LL(1)? (Select all that apply. Each option is considered independently, i.e., exactly one production is added to the grammar at a time.)

A. $C \rightarrow x$

B. $C \rightarrow y$

C. $C \rightarrow zx$

D. $C \rightarrow \epsilon$

Your Answer

Not answered

Correct Answer

B. Option B, C. Option C, D. Option D

Option A

Option A

Option B

Option B

CORRECT

Option C

Option C

CORRECT

Option D

Option D

CORRECT

OFFICIAL SOLUTION

Answer: B, C, D

[➤ Official discussion page](#)


Q62 | Technical

UNATTEMPTED

The following grammar is ambiguous:

$$A ::= B b C$$

$$B ::= b \mid \epsilon$$

$$C ::= b \mid \epsilon$$

To demonstrate this ambiguity we can use pairs of derivations. Here are four different pairs. For each pair of derivations, select it if the pair correctly proves that the grammar is ambiguous. Do not select if the pair does not give a correct proof.

Which of the following pairs of derivations correctly prove that the grammar is ambiguous?

Your Answer

Not answered

Correct Answer

B. $\&A \Rightarrow BbC \Rightarrow bbC \Rightarrow bb$
 $\&A \Rightarrow BbC \Rightarrow bC \Rightarrow bb$ ^{

}

Option A

$\&A \Rightarrow BbC \Rightarrow bbC \Rightarrow bbb$

$\&A \Rightarrow BbC \Rightarrow Bbb \Rightarrow bbb$

Option B

$\&A \Rightarrow BbC \Rightarrow bbC \Rightarrow bb$

$\&A \Rightarrow BbC \Rightarrow bC \Rightarrow bb$

CORRECT

Option C

$\&A \Rightarrow BbC \Rightarrow bbC \Rightarrow bb$

$\&A \Rightarrow BbC \Rightarrow Bbb \Rightarrow bb$

Option D

$\&A \Rightarrow BbC \Rightarrow bbC \Rightarrow bb$

$\&A \Rightarrow BbC \Rightarrow bbC \Rightarrow bbb$

OFFICIAL SOLUTION

Answer: B

[Official discussion page](#)



Q63 | Technical

UNATTEMPTED

Refer to the given 3-address code sequence. This code sequence is split into basic blocks. Let the number of basic blocks be **N**.

1: $i = 1$

2: $b = i + 2$

```

3: y = i + b
4: L1: if i > 20 then goto L2
5:     print b+y
6: L2: b = b+1
7:     if b > 10 then goto L3
8:     i = i+1
9:     y = b+1
10:    goto L1
11: L3: return y

```

Draw the control flow graph (CFG) **G** for the above program.

In the context of liveness analysis, liveness is associated with edges.

Let **M** be the number of edges in **G** where each of the variables **b**, **i**, and **y** are live.

What is **M + N** ?

Your Answer

Not answered

Correct Answer

11

Accepted Answer

11

OFFICIAL SOLUTION

Answer: Not available

[↗ Official discussion page](#)

Q64 | Technical

UNATTEMPTED



The following is the body of an entire procedure that uses six temporaries

\verb|a|, **\verb|b|**, **\verb|c|**, **\verb|d|**, **\verb|e|**, **\verb|f|**.

```

a = 1
b = 2
c = a + b
d = a + c
e = c + d
f = d + 3
b = e + 4
c = b + e
f = b + f
b = c + f
return b + 5

```

The following fragment of the above procedure uses four temporaries

\verb|a|, **\verb|b|**, **\verb|c|**, **\verb|d|**.

```

b = 2
c = a + b

```

$$d = a + c$$

Assume that we have done register allocation for the entire procedure that contains this program fragment and that we have assigned **a** to register **r1**, **b** to register **r2**, **c** to a memory location that we denote by **c** and **d** to a memory location that we denote by **d**. Assume also that we have a spare register **v0**.

For a load/store architecture, in which only load and store instructions can have memory operands and arithmetic instructions can have only register or immediate operands, what is a correct code for the above program fragment?

Your Answer

Not answered

Correct Answer

```

&move r2 := 2
&move v0 := r1 + r2
B.      &store c := v0<br><br>
&move v0 := r1 + v0
&store d := v0

```

Option A

```

& move r2 := 2
& move v0 := r1 + r2
& store c := v0
& move v0 := r1 + r2
& store d := v0

```

Option B

```

&move r2 := 2
&move v0 := r1 + r2
&store c := v0
&move v0 := r1 + v0
&store d := v0

```

CORRECT**Option C**

```

&move r2 := 2
&move v0 := r1 + r2
&store c := r1
&move v0 := r1 + v0
&store d := v0

```

Option D

```

&move r2 := 2
&move v0 := r1 + r2
&store c := v0
&move v0 := r1 + v0
&store d := r2

```

OFFICIAL SOLUTION

Answer: B

[Official discussion page](#)

Q65 | Technical

UNATTEMPTED

Consider the grammar:

$$A ::= yB \mid xyB$$

$$B ::= xy \mid xA$$

where $\{A, B\}$ is the set of non-terminal symbols, A is the start symbol, $\{x, y\}$ is the set of terminal symbols, and ϵ denotes the empty string.

Which grammar generates the same language as the above grammar?

Your Answer

Not answered

Correct Answer

$$\begin{aligned} A &::= yB \mid xyB \\ B &::= xC \\ \text{A. } C &::= yD \mid xyB \\ D &::= \epsilon \mid B \end{aligned}$$

Option A

CORRECT

$$A \& ::= yB \mid xyB$$

$$B \& ::= xC$$

$$C \& ::= yD \mid xyB$$

$$D \& ::= \epsilon \mid B$$

Option B

$$\&A ::= yB \mid xyB$$

$$\&B ::= xy \mid xyB \mid xxA$$

Option C

$$\&A ::= yxyC \mid xyC$$

$$\&B ::= xC$$

$$\&C ::= yD \mid xyD$$

$$\&D ::= \epsilon \mid B$$

Option D

None of the other cases.

OFFICIAL SOLUTION

Answer: A

[↗ Official discussion page](#)

